

MRI: What, Why and When?



“Without history human is demoted to lower animals”

- ▶ Dr Isidor Rabi (Nobel in 1944), discovered NMR (Nuclear Magnetic Resonance) in the late 1930s, but considered it to be an artefact of his apparatus!
- ▶ Bloch and Purcell were awarded the Nobel Prize for Physics in 1952 for the discovery of NMR, and is widely used in assessing complex chemical compounds.

We are so close to the man behind MRI

- ▶ Prof Peter Mansfield was awarded Nobel in 2003 for his discoveries in MRI (with Prof Paul C Lauterbur of USA)
- ▶ Peter Mansfield is from Nottingham University, UK



MRI: What is it?

Magnetic Resonance Imaging

Magnet

Radio Frequency

Imaging

WE ARE MAGNETS!

Really?



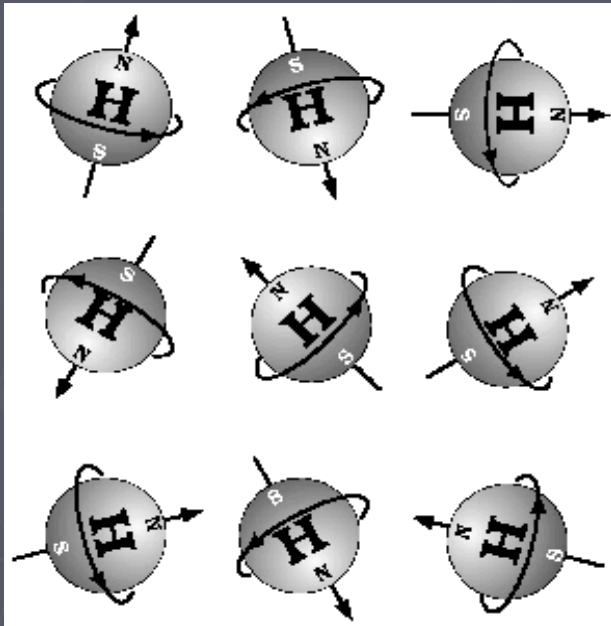
We all are made up of elements

- ▶ 92 elements occur naturally on earth.
- ▶ Human body is built of only 26 elements.
- ▶ Oxygen, hydrogen, carbon, nitrogen elements constitute 96 % of human body mass.
- ▶ Oxygen is 65 % of body mass; carbon is 18.5 %, hydrogen 9.5 %, nitrogen 3.2 %.
- ▶ Let us ignore all elements but Hydrogen.

Why hydrogen?

- ▶ Simplest element with atomic number of 1 and atomic weight of 1
- ▶ When in ionic state (H^+), it is nothing but a proton.
- ▶ Proton is not only positively charged, but also has magnetic spin (wobble)!
- ▶ MRI utilizes this magnetic spin property of protons of hydrogen to elicit images!!
- ▶ We are magnets!

But why we can't act like magnets?

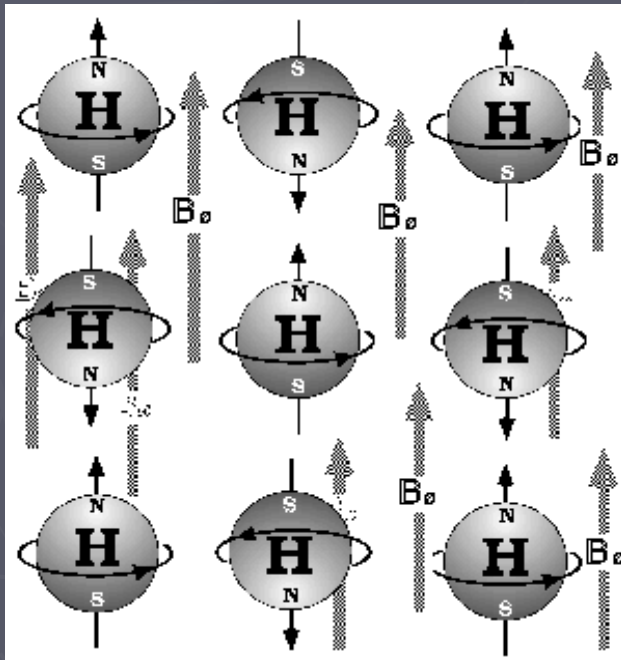


- ▶ The protons (i.e. Hydrogen ions) in body are spinning in a haphazard fashion, and cancel all the magnetism. That is our natural state!
- ▶ We need to discipline them first, how?

We need a big magnet from outside!

- ▶ Magnetic field strength: 0.3 – 7 T (2500 times more than earth's magnetic field). Average field strength – 1.5 T
- ▶ Open magnet – less field strength, less claustrophobic
- ▶ Closed magnet – more field strength, claustrophobic

Proton alignment



- ▶ Compass aligns with the earth
- ▶ In a similar fashion,
- ▶ Our body protons (hydrogen) align with this external magnetic field.
- ▶ Now, we are disciplined (spinning in line with each other!), what next?

Now, its time to listen to radio in RESONANCE.

- ▶ Pushing a swing in time with natural interval of the swing will make the swing higher and higher.
- ▶ Similarly, radio frequency pulses in resonance push the aligned protons (H^+) to a higher energy level.

What is Radio Frequency pulse?

- ▶ Same as Radio waves – high wavelength, low energy electromagnetic waves
- ▶ Radiofrequency coils
 - Act as transmitter and receiver
 - Different types of coils

Turn off the radio

- ▶ The higher energy gained by the protons is retransmitted (NMR signal)
- ▶ The original magnetization begins to recover (T1)
- ▶ The excessive spin begins to dephase (T2)

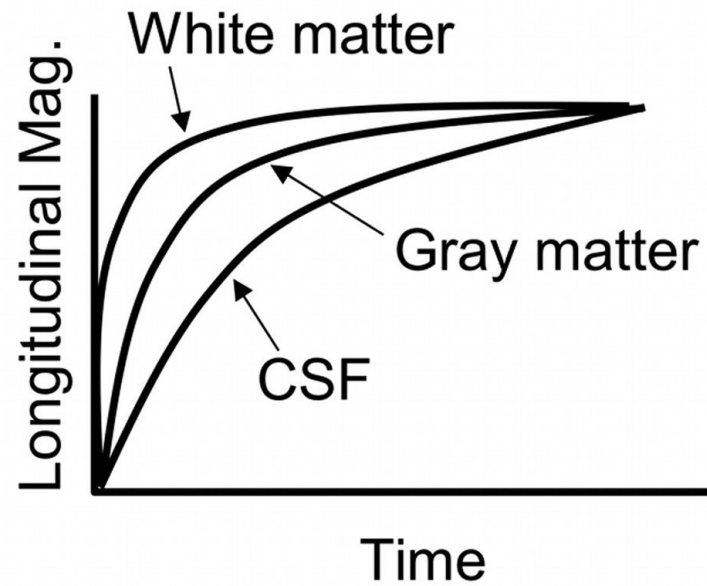
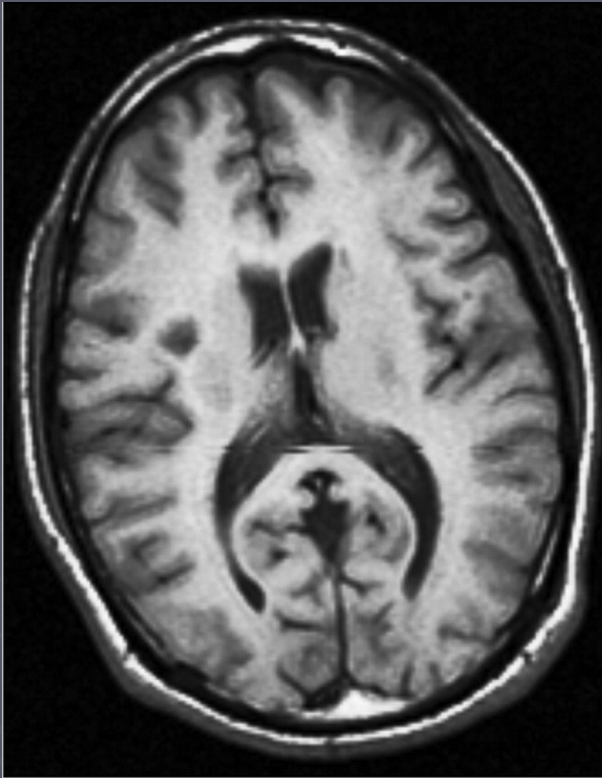
Now, we re-transmit the energy for image processing

- ▶ The emitted energy is too small (despite 2500 times the magnetic field with resonance RF pulse) to convert them into images.
- ▶ Hence, repeated “ON-OFF” of RF pulses are required.
- ▶ The emitted energy is stored (K-space), analysed and converted into images.
- ▶ What kind of images?

What kind of images?

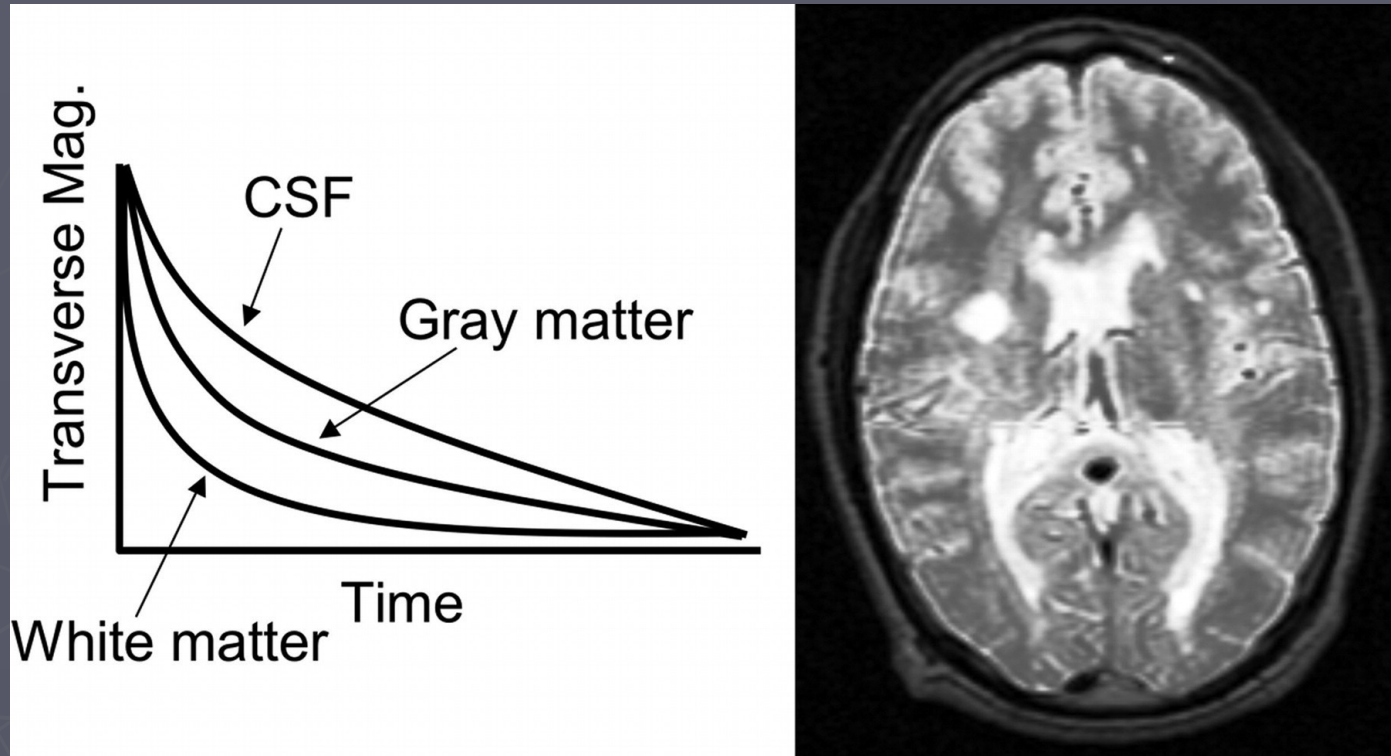
- ▶ T1WI
- ▶ T2WI
- ▶ PDWI
- ▶ DWI
- ▶ ADC
- ▶ GE
- ▶ Perfusion images
- ▶ fMRI
- ▶ BOLD images
- ▶ MRA
- ▶ MRV
- ▶ Post-Gd images
- ▶ Volumetric images
- ▶ MR arthrograms
- ▶ FLAIR
- ▶ STIR
- ▶ Etc etc etc

T1-weighted contrast



Pooley, R. A. Radiographics 2005;25:1087-1099

T2-weighted contrast



Pooley, R. A. Radiographics 2005;25:1087-1099

Tissue Appearance

WT	FAT	H2O	MUSC	LIG	BONE
T1	B	D	I	D	D
Proton	I	I	I	D	D
Density T2	I	B	I	D	D

T1WI

	Air, mineral rich tissue (cortical bone, stones), fast-flowing blood
Dark	
	Collagenous tissue (ligaments, tendons, scars), high free water tissue (kidneys, gonads, edema, fluids [urine, bile], simple cysts, bladder, gallbladder, spleen, CSF), high bound water tissues (liver, pancreas, adrenals, hyaline cartilage, muscle)
Low	
	Proteinaceous tissue (abscess, complex cysts, synovial fluid)
Intermediate	
	Fat, fatty bone marrow, blood products (methemoglobin [metHb]), slow-flowing blood, radiation change, paramagnetic contrast agents
Bright	

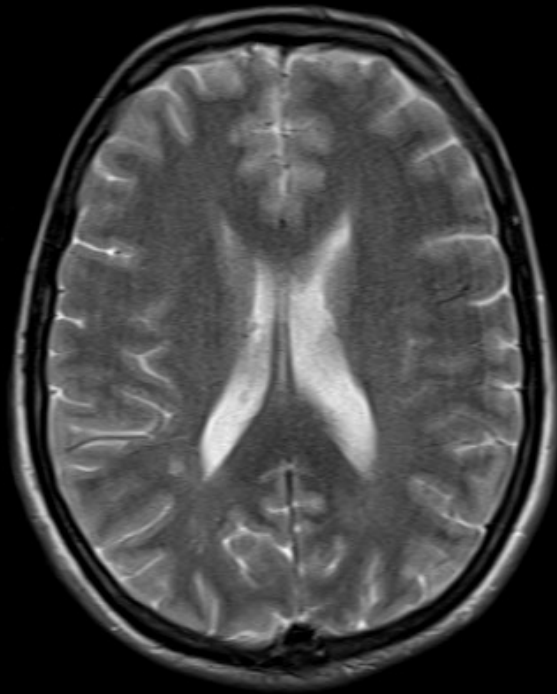
T2WI

	Air, mineral rich tissue (cortical bone, stones), fast-flowing blood
Dark	
	Collagenous tissue (ligaments, tendons, scars), bone islands
Low	
	High bound water tissues (liver, pancreas, adrenals, hyaline cartilage, muscle)
Intermediate	
	Fat, fatty bone marrow
Bright	
	High free water tissue (kidneys, gonads, edema, fluids (urine, bile), simple cysts, bladder, gallbladder, spleen, CSF), proteinaceous tissue, blood products (oxyhemoglobin, extracellular metHb)

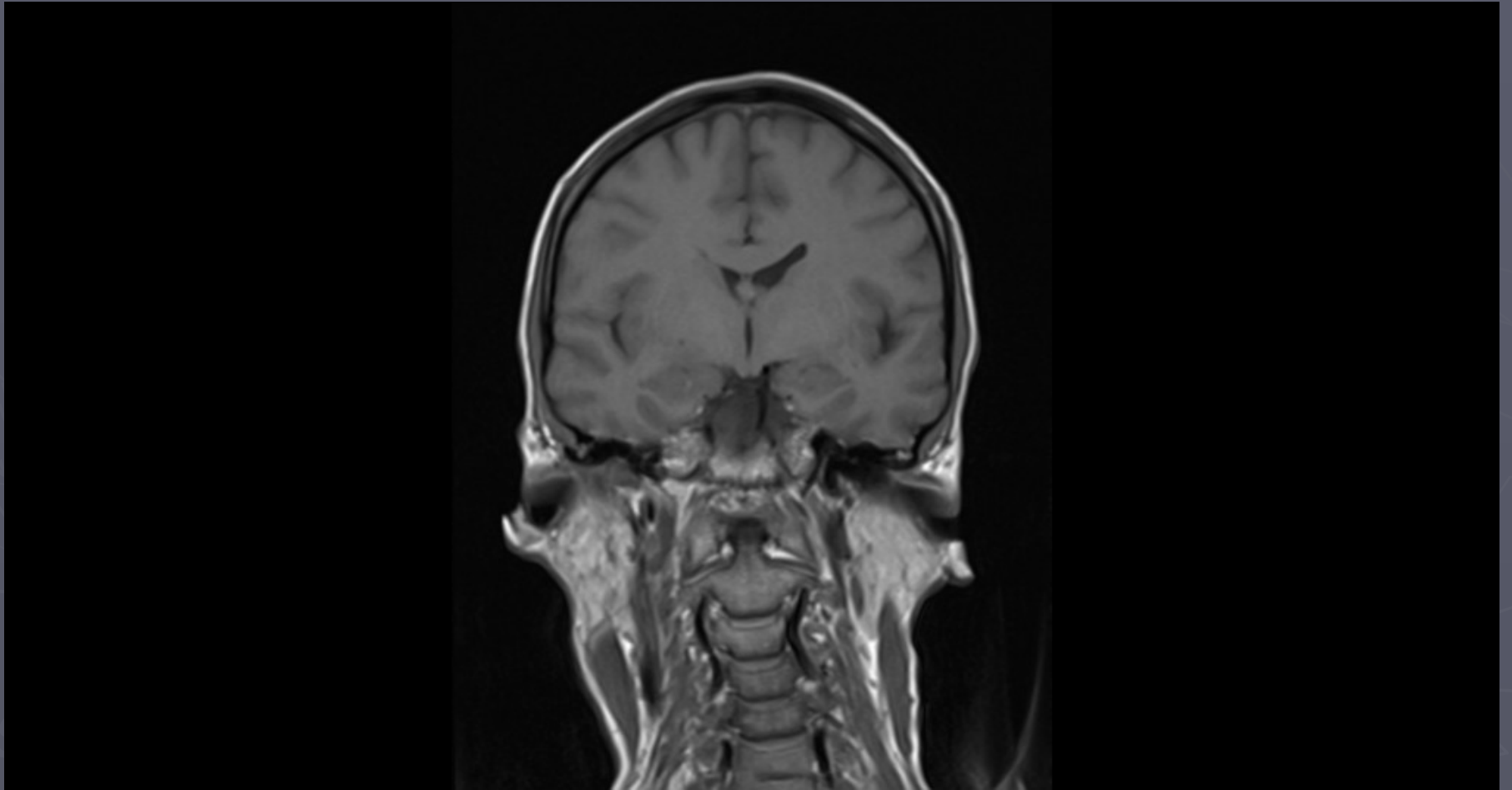
A bit of sequence exercise



Which sequence is it? T2



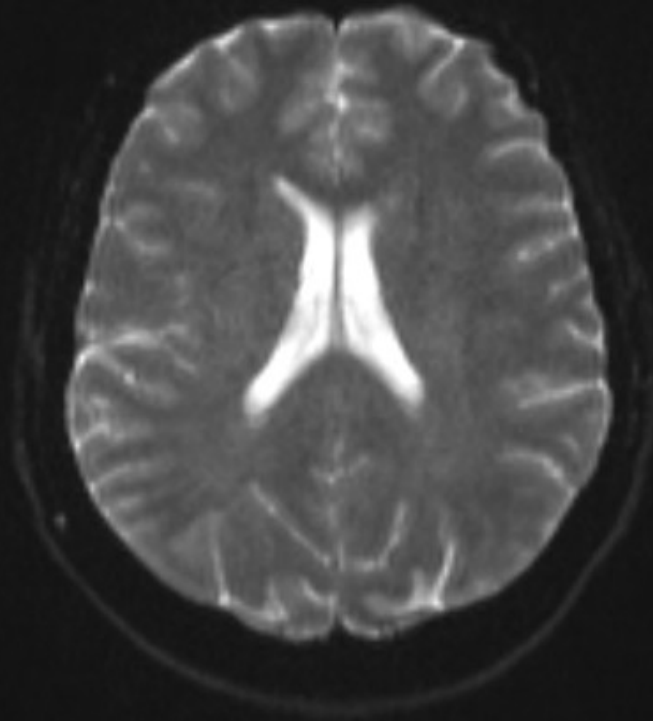
Which sequence is it? T1



Which sequence is it? FLAIR



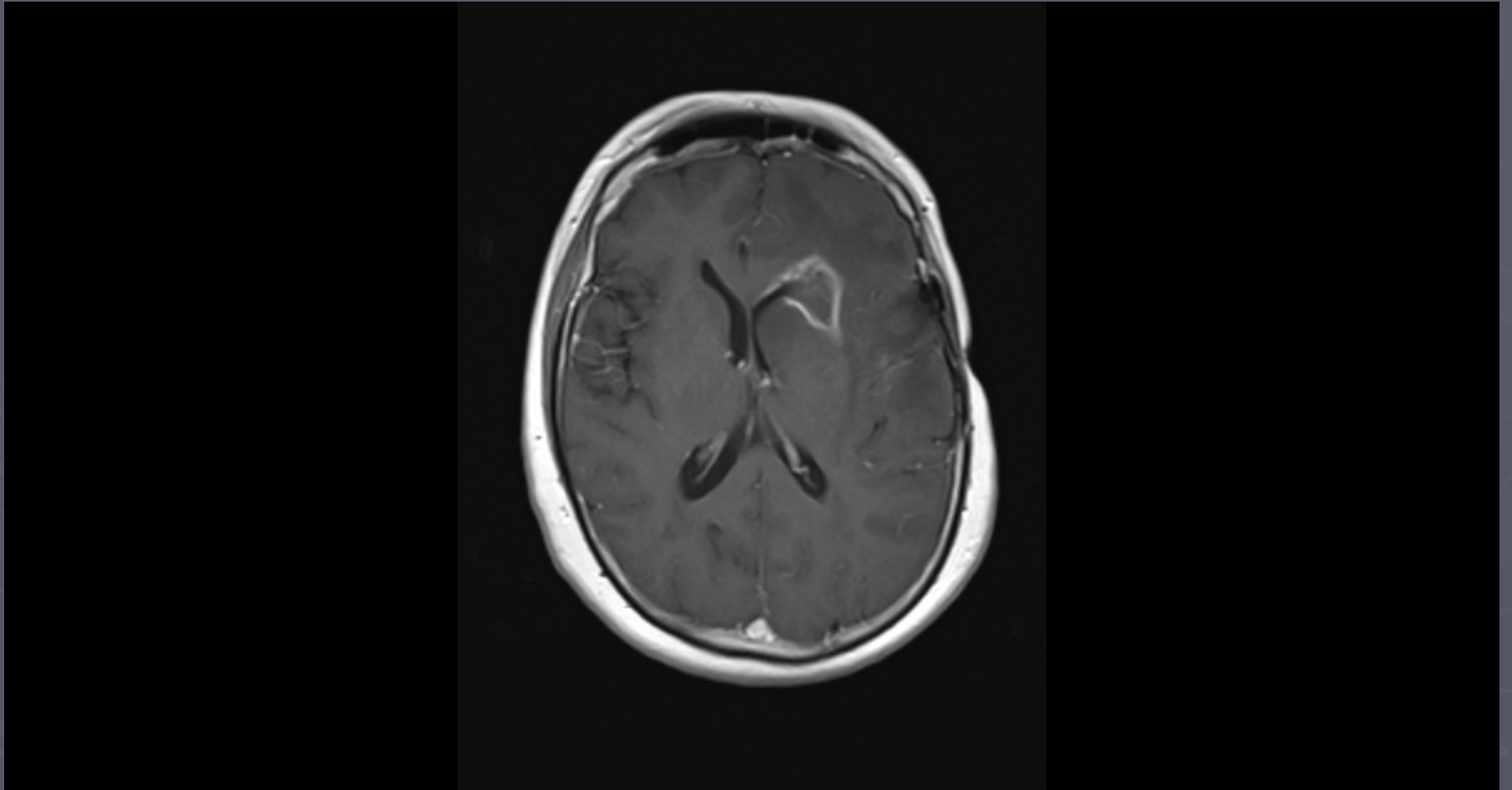
Which sequence is it? DWI



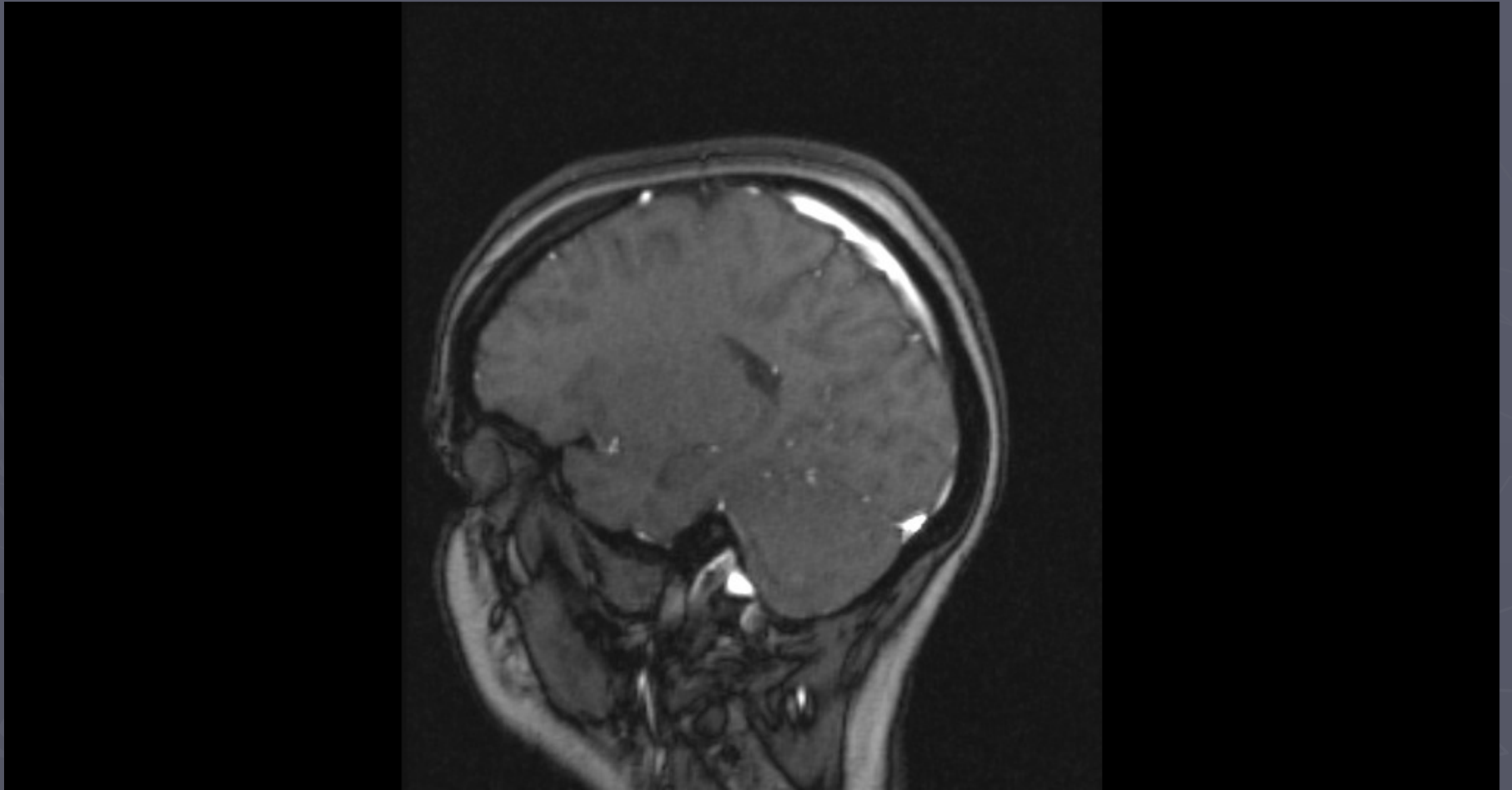
Which sequence is it? ADC



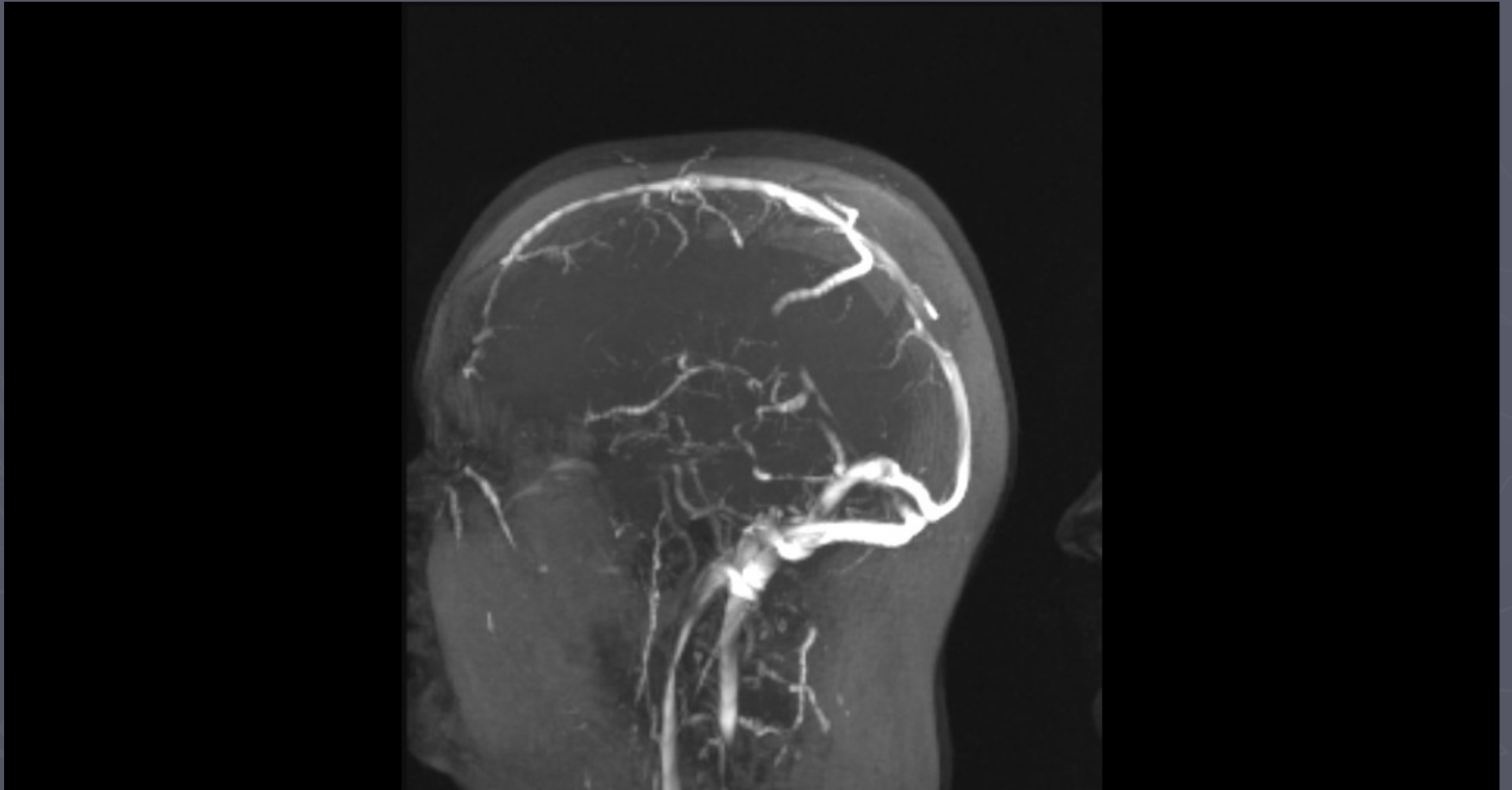
Which sequence is it? PostGd



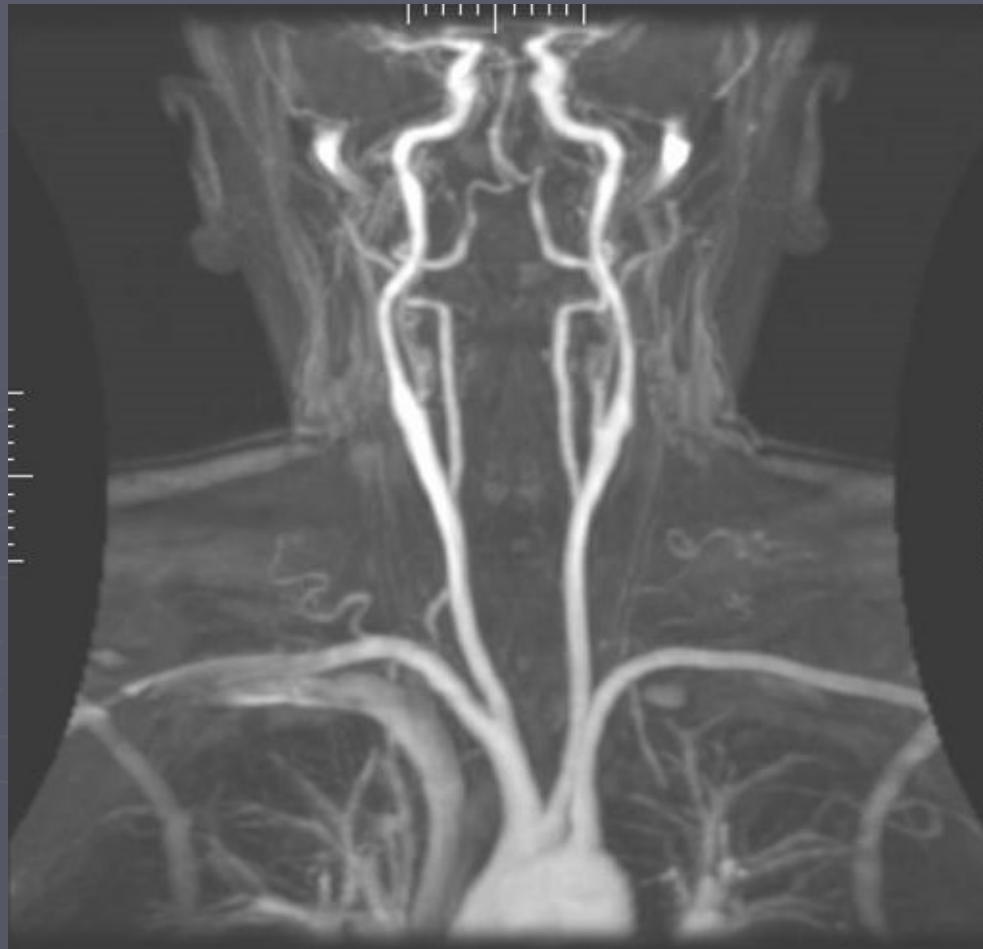
Which sequence is it? MRV



Which sequence is it?, 3D MRV



Which sequence is it? MRA

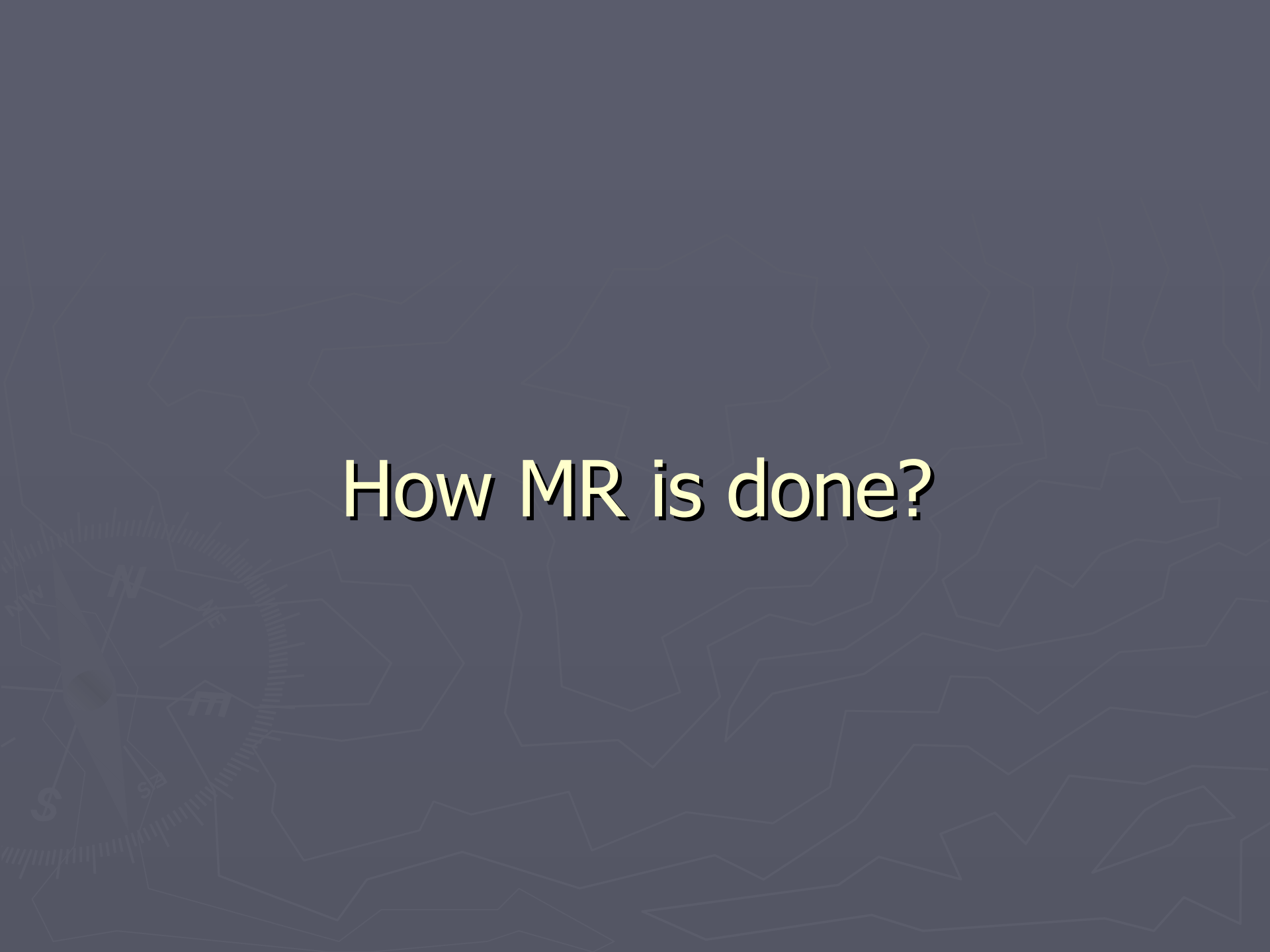


Which sequence is it? T2



Which sequence is it? T1



The background of the slide is a dark blue-grey color. It features a faint, light-grey graphic on the left side that combines a compass rose with a topographic map. The compass rose shows cardinal and intercardinal directions (N, NE, E, SE, S, SW, W, NW) and degree markings. The topographic map consists of irregular, wavy lines representing contour lines. The text "How MR is done?" is centered in a bold, yellow, sans-serif font with a black drop shadow.

How MR is done?

In a Tunnel, of course!



Before entering tunnel, there is a checklist!



- ▶ No mobiles, no credit cards, please!
- ▶ Known potential safety concerns due to large static magnetic field:
 - Internal cardiac pacemakers
 - Steel cerebral aneurysm clips (ferromagnetic)
 - Small steel slivers embedded in eye
 - Life-support equipment with magnetic steel
 - Cochlear implants
 - Stents anywhere in the body

Is entering the tunnel safe?

- ▶ No definite long-term harmful effects
- ▶ Pregnancy is a relative contraindication, as we will never be able to tell with 100% certainty that MRI is 100% safe during pregnancy!
- ▶ Babies and children need sedation or GA
- ▶ Some people fear tunnels (claustrophobia)

What happens in MRI?

- ▶ Stay still for 15 minutes to 45 minutes!
- ▶ Noise, Noise and Noise!
- ▶ Listen to music in darkness
- ▶ Alien (radiographers) like voices in between, “another 5 minutes to go”, or, “please stay still”.
- ▶ Somebody can come and inject.

Clinical Aspects of MRI



When to MRI?

- ▶ When everything else fails, there is MRI
- ▶ When you want to borrow time from the patient, request MRI and hope that the waiting time will take care of the patient

Advantages of MRI

1. No ionizing radiation & no short/long-term effects demonstrated
2. Variable thickness, any plane
3. Better contrast resolution & tissue discrimination
4. Various sequences to play with to characterise the abnormal tissue
5. Many details without I.V contrast

Disadvantages of MRI

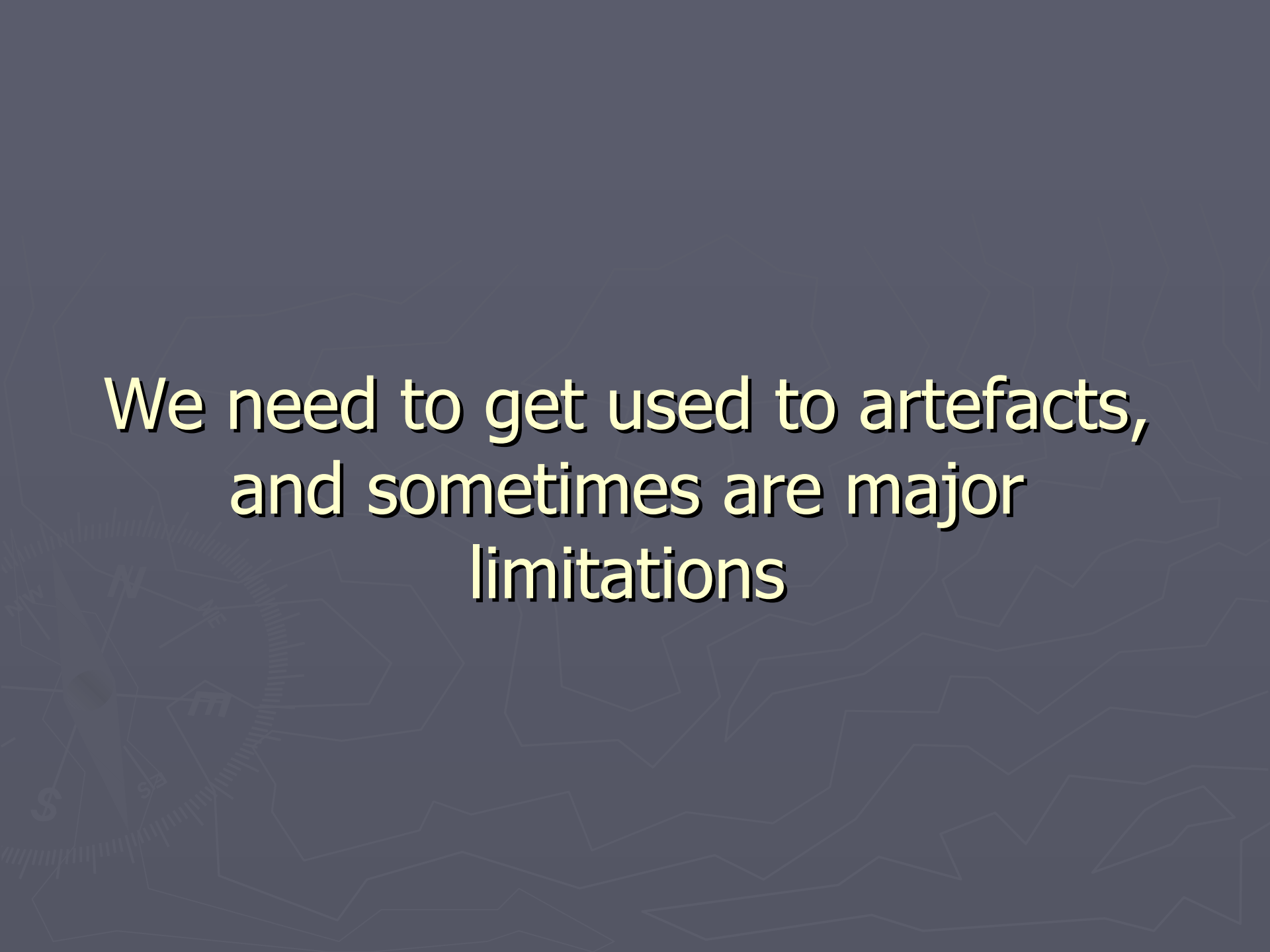
- ▶ Time consuming
- ▶ Not easily available (long waiting list)
- ▶ No on-call service
- ▶ Need to tweak sequences as per the clinical questions; hence cannot be generalised
 - Pain abdomen - ? cause

Nothing is perfect in this world

- ▶ MRI has limitations:
 - Bone
 - Air
 - Time consuming
 - Poor spatial resolution
 - Expertise!

We presumed MR contrast is safe

- ▶ No side effects
- ▶ No allergy (as with Iodine)
- ▶ Can be used in renal impairment
- ▶ Can be used as CT contrast when a patient has impaired renal functions!

The background of the slide features a dark blue-grey color. On the left side, there is a faint, light-grey graphic of a compass rose with a needle pointing towards the top-left. To the right of the compass, there is a faint, light-grey topographic map showing various contour lines and shapes.

We need to get used to artefacts,
and sometimes are major
limitations

How to get the best from MRI?

- ▶ Ask a specific question
- ▶ Get a specific answer
- ▶ Because the sequences can be tailored accordingly.

CT

- ▶ Faster
- ▶ Less expensive
- ▶ Less sensitive to patient movements
- ▶ Easier in claustrophobics
- ▶ Acute haemorrhage
- ▶ Calcification
- ▶ Bone details
- ▶ Foreign body

MR

- ▶ No ionising radiation
- ▶ Greater details, hence more sensitive and more specific
- ▶ Any plane scanning
- ▶ Contrast less allergic
- ▶ No beam hardening artefact

Neuroimaging = MRI

- ▶ No neurology or neurosurgery without MRI
- ▶ MR brain has largely replaced CT brain in USA (but for head injury, suspected acute intracranial haemorrhage)
- ▶ Superior to CT in most occasions
- ▶ CT is poor man's MRI in all other indications!
- ▶ Show me a person with migraine who has not undergone MRI in USA!

Indications (almost everything)

- ▶ Anatomy
- ▶ Congenital anomalies
- ▶ Hereditary and metabolic diseases
- ▶ Infections
- ▶ Demyelination
- ▶ Vascular event
- ▶ Tumours
- ▶ Trauma
- ▶ Dementia
- ▶ Hydrocephalus
- ▶ Cranial nerves
- ▶ Arteriograms
- ▶ Venograms
- ▶ Skull base
- ▶ Pituitary gland

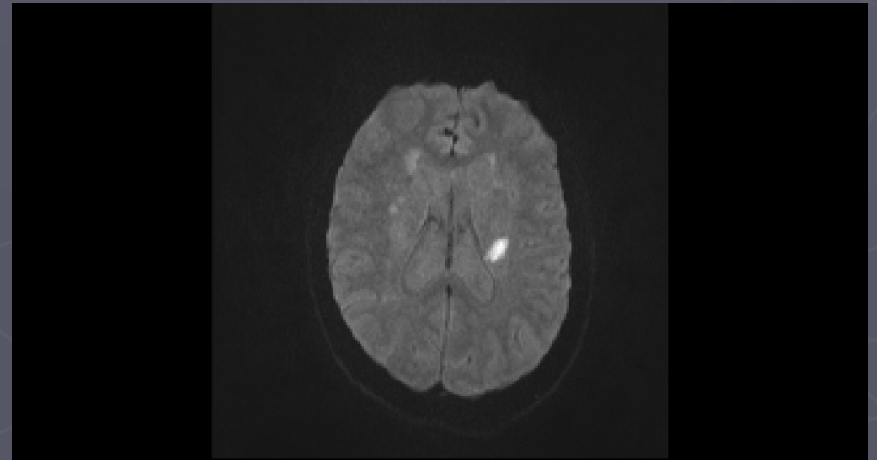
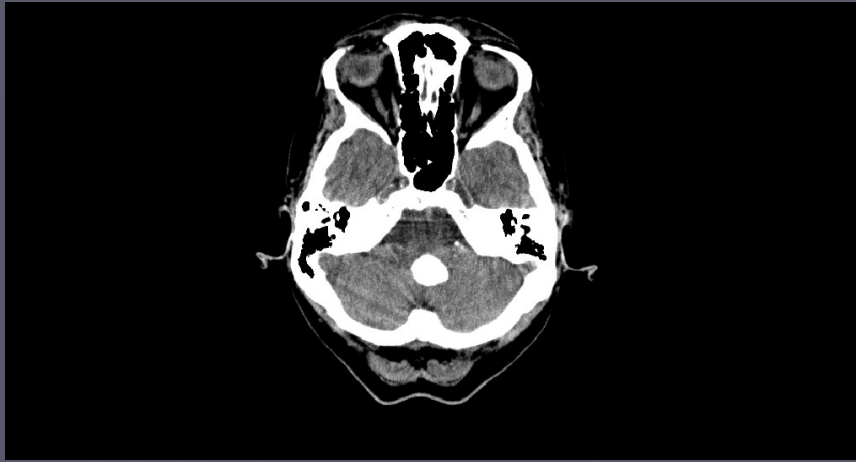
Tools in neuroimaging

- ▶ T2WI
- ▶ T1WI
- ▶ FLAIR
- ▶ MRA
- ▶ MRV
- ▶ GE
- ▶ Gadolinium
- ▶ DWI
- ▶ ADC
- ▶ Diffusion tensor imaging
- ▶ fMRI
- ▶ Perfusion mapping
- ▶ CSF flow mapping
- ▶ MR spectroscopy

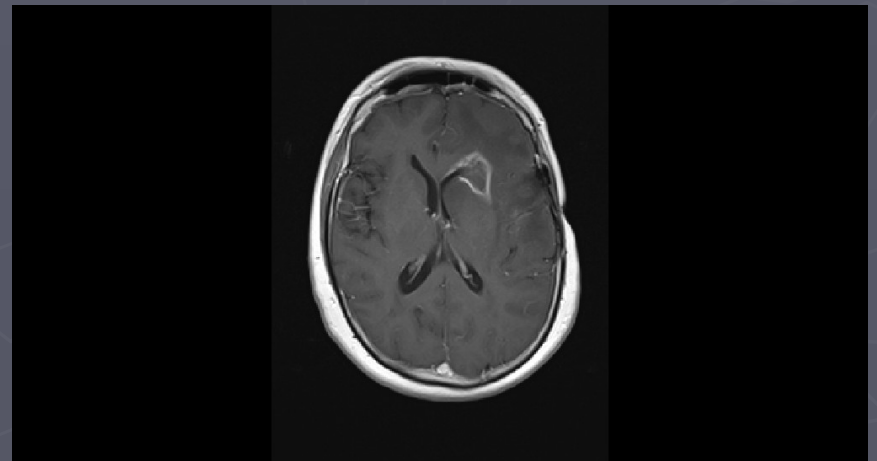
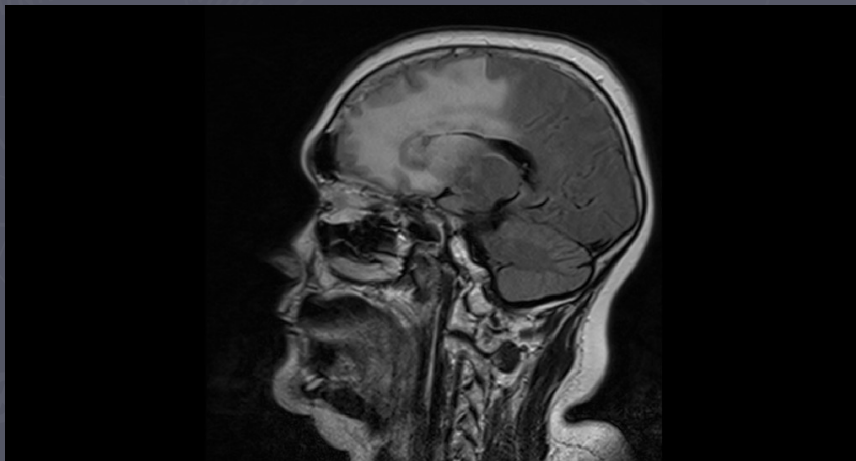
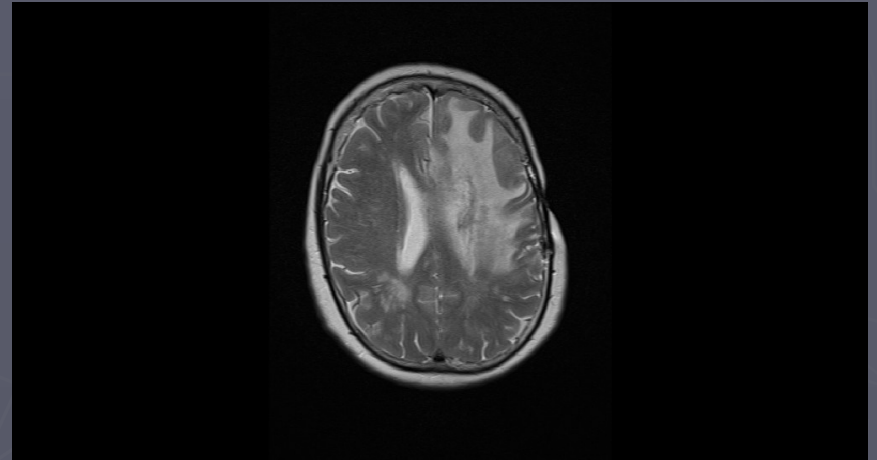
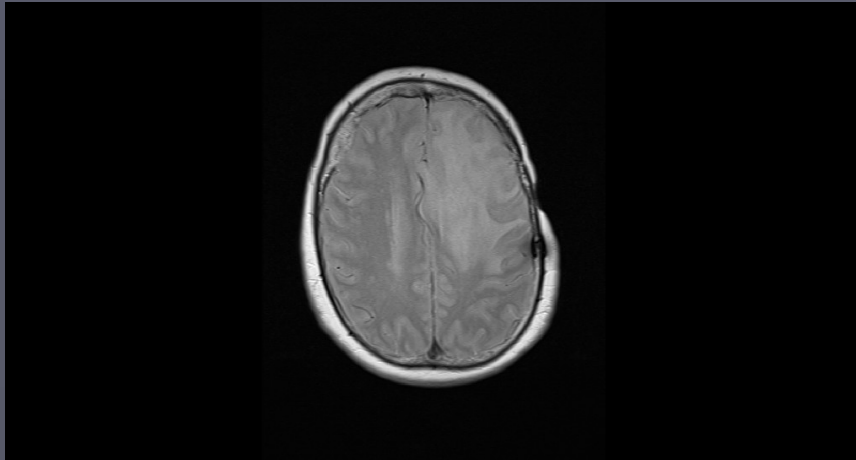
Stroke imaging

- ▶ MR superior to CT in diagnosing hyperacute infarct
- ▶ MR is as sensitive as CT in diagnosing acute intracranial haemorrhage
- ▶ MR is more expensive and less easily available compared to CT in the UK
- ▶ CT is currently widely used to exclude haemorrhage before thrombolysis

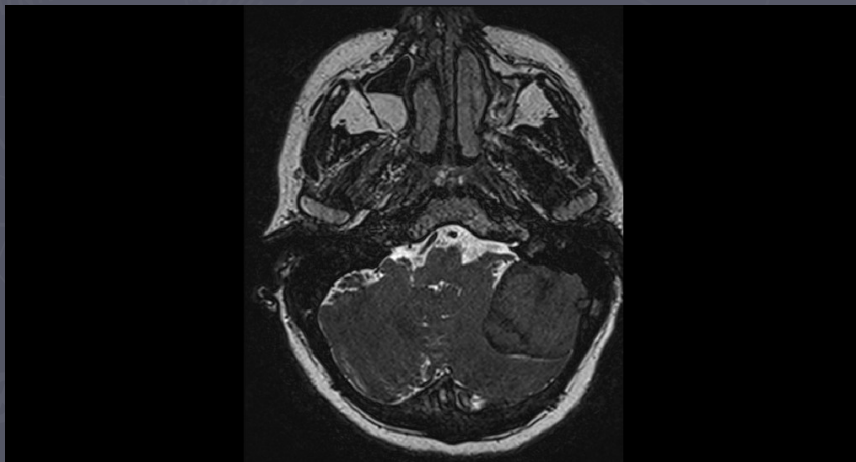
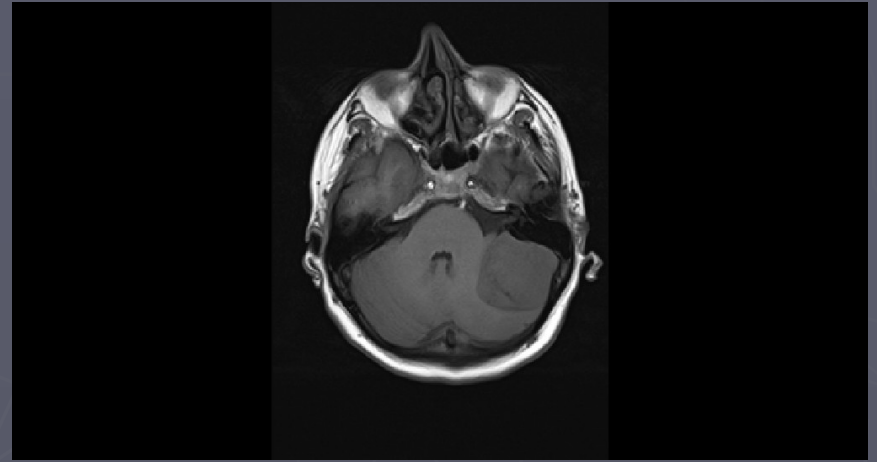
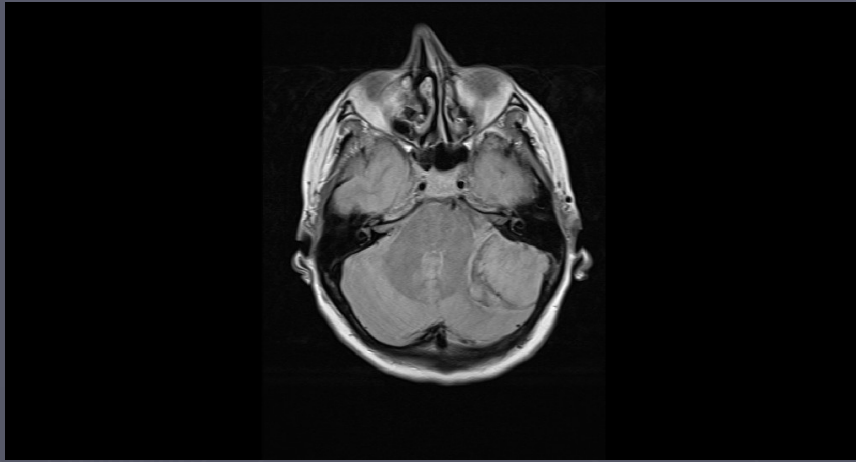
Acute infarct best shown on DWI



glioma



meningioma



Spine imaging

- ▶ MR is the investigation of choice
- ▶ Conventional CT, CT myelogram and conventional myelogram are no longer performed, unless MR is contraindicated.
- ▶ Indications and contraindications – same
- ▶ First line of investigation in suspected spinal infection, cord compression, cauda equina, sciatica

Disc lesion and MRI

- ▶ Virtually everyone after the age of 40 years will have at least one degenerative disc/ end plate
- ▶ Not all patients with sciatica will have a positive MRI
- ▶ Ask MRI to answer a specific question; otherwise MRI might completely mislead the clinician.

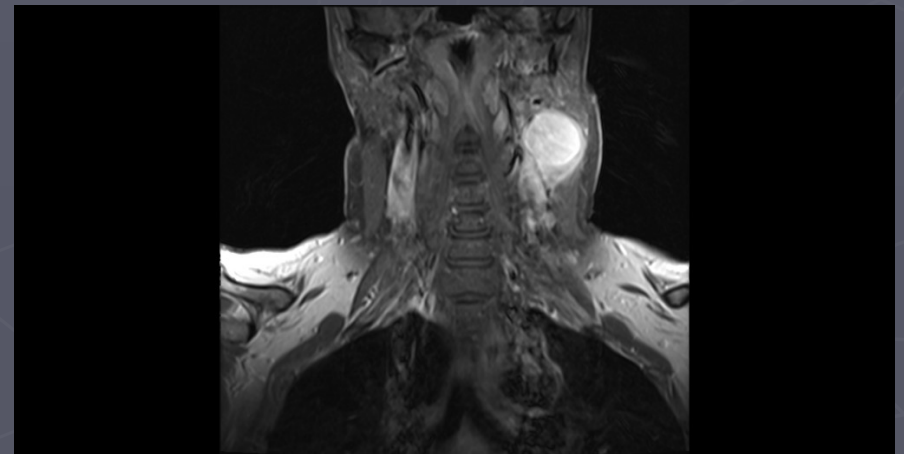
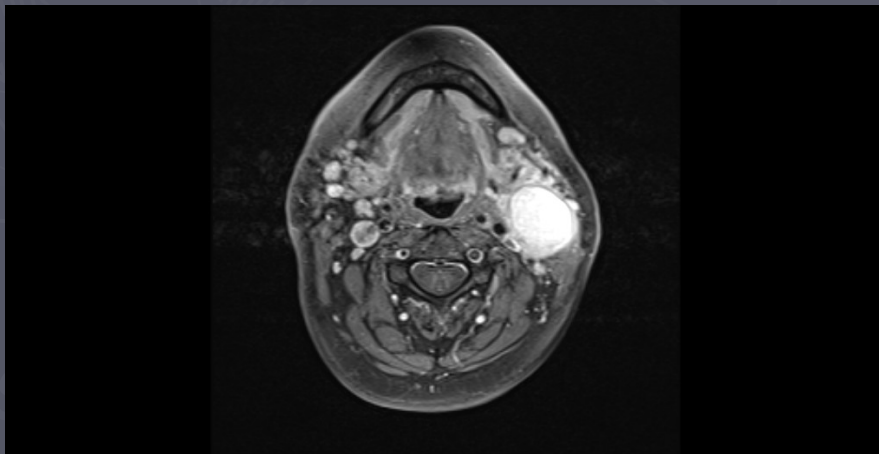
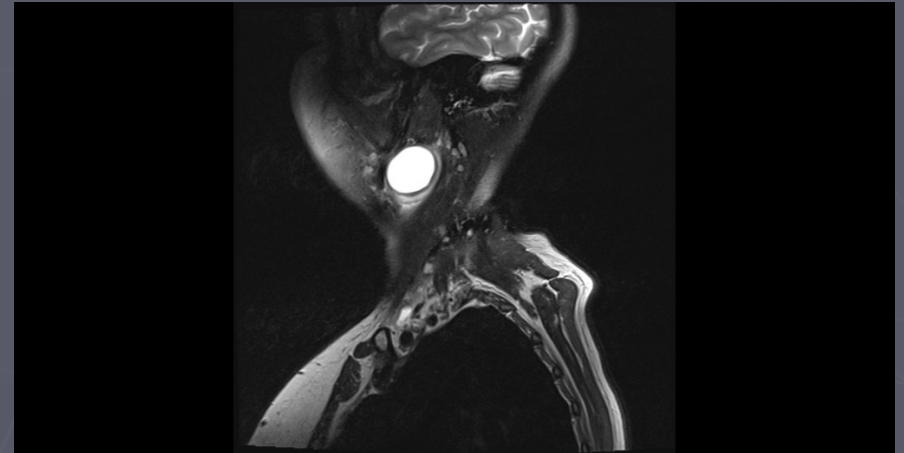
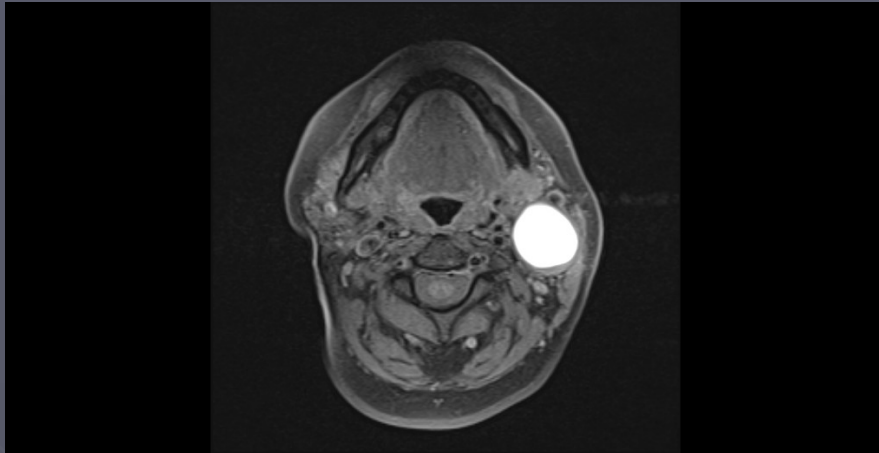
Metastasis



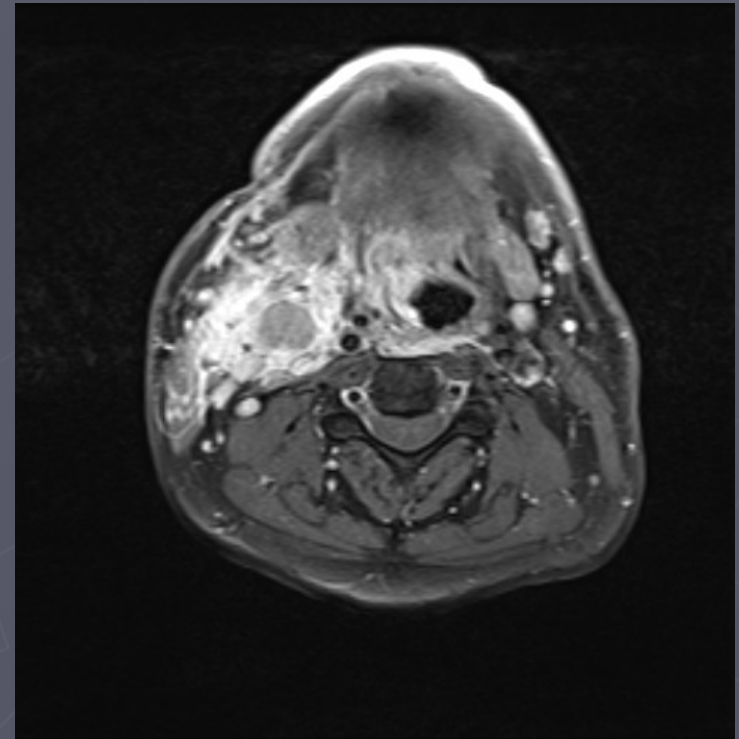
Head and Neck imaging

- ▶ MR is complementary or second line of investigation in many of head and neck pathologies
- ▶ Superior to CT in staging head and neck malignancies
- ▶ Characterise the head and neck lesions better than CT
- ▶ Complementary to CT in petrous temporal and paranasal sinus evaluation
- ▶ First line of investigation in orbital lesions

Second branchial cleft cyst



Tongue base carcinoma



Musculoskeletal imaging

- ▶ Initial evaluation of bones – Plain films; then MRI
- ▶ MRI sensitive than CT in diagnosing occult fractures
- ▶ Initial evaluation of soft tissues – USG; then MRI
- ▶ Joint imaging = MRI

Tools in MSK imaging

- ▶ T1WI
- ▶ T2WI
- ▶ Fat sat T1
- ▶ STIR
- ▶ Fat sat T2
- ▶ Gadolinium studies
- ▶ MR arthrography

Indications

- ▶ Occult fractures
- ▶ Marrow abnormality
- ▶ Ligament pathologies
- ▶ Tendon pathologies
- ▶ Muscular injuries
- ▶ Infection
- ▶ Bone and soft tissue tumour
- ▶ Labral pathologies

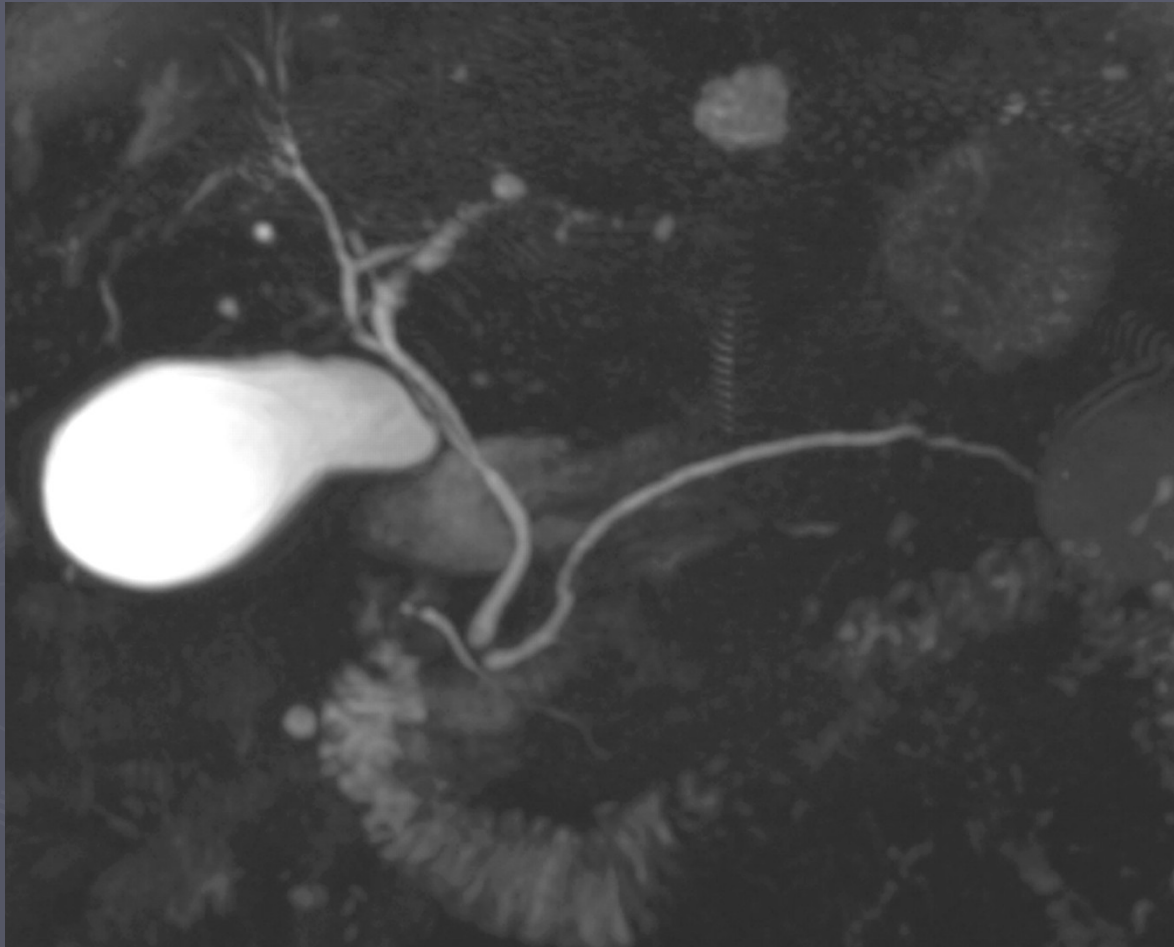
Fracture



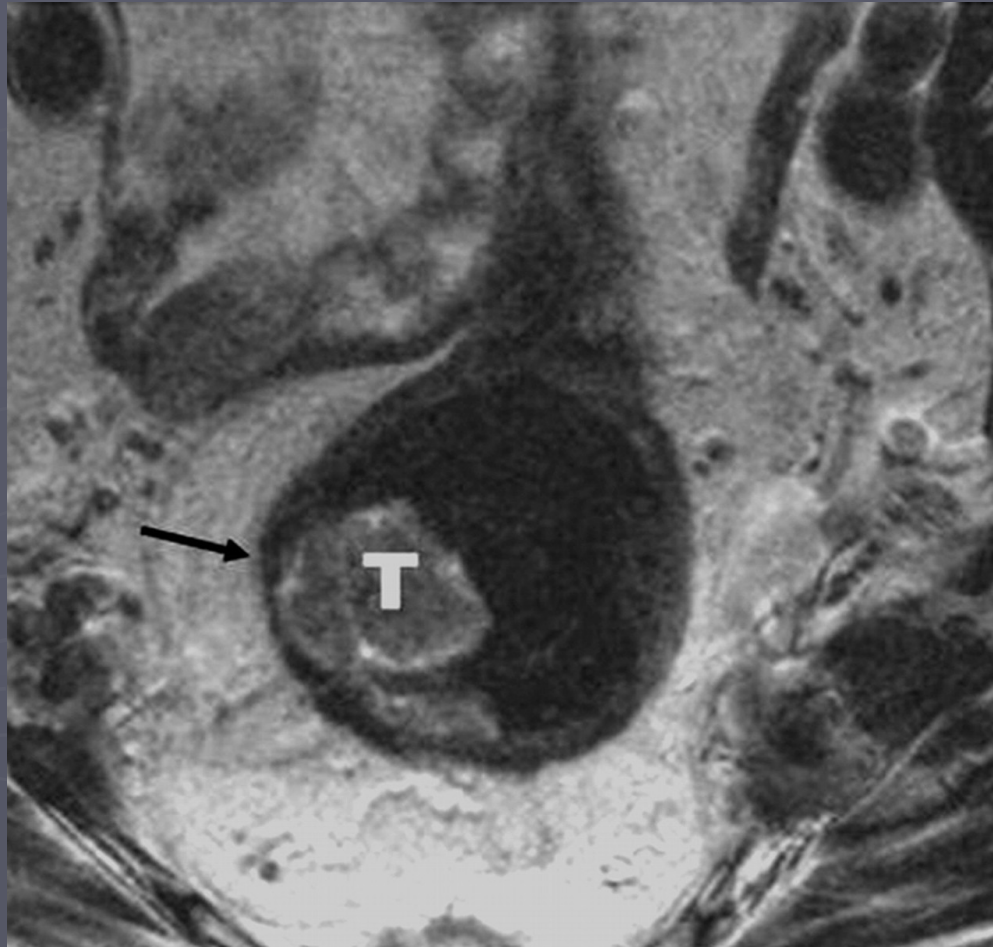
Abdominal MRI

- ▶ Problem solving tool in liver, pancreatic, renal and adrenal lesions
- ▶ Primary modality in local staging of rectal ca, endometrial ca, cervical ca, prostate ca, vaginal ca
- ▶ Non-invasive modality in evaluating pancreaticobiliary tract – MRCP
- ▶ Scrotal and penile imaging
- ▶ Uterus and ovary imaging

MRCP



Rectal ca



Vascular MRI

- ▶ Peripheral vascular arteriogram with or without I.V contrast
- ▶ Aortogram
 - Dissection
- ▶ Pulmonary arteriogram
 - When CT is contraindicated

Cardiac MRI

- ▶ Coming in a big way
- ▶ Very useful in congenital heart diseases, cardiomyopathies
- ▶ Evidence is emerging in the evaluation of myocardial infarction

Breast MRI

- ▶ Problem solving
 - Breast implants
 - Recurrence
 - Multifocal disease

Summary

- ▶ Expensive time-consuming investigation
- ▶ Complex physics, too many sequences, difficult to interpret to untrained eyes
- ▶ Relatively safe, but there are definite contraindications
- ▶ Ask specific question to get the right answer

Summary

- ▶ MRI invaluable imaging tool in the diagnosis of various diseases from head to toe
- ▶ Chief modality in neuroimaging, and musculoskeletal imaging
- ▶ Problem solving tool in abdominal pathologies
- ▶ Invaluable tool in local staging of most of the malignancies